

Control of a UAV Group Using Nonlocal Virtual Potential (Verlet Algorithm)

Abstract

A mathematical model of motion of a UAV group on a plane inside a circle is presented. Based on a nonlocal potential, the repulsion of the aircraft from the circle boundary and the absence of collisions between the aircraft are taken into account. The numerical implementation of the group dynamics is based on the Verlet method. The numerical integration method is tested using test problems. Methods for calculating the dynamics of a UAV group in a circle with different initial conditions and number of aircraft are illustrated.

Introduction

UAV groups are an effective way to perform various tasks for economic and military purposes. The use of aircraft-type UAV groups requires highly qualified software and expensive equipment. In this case, achieving the final mission goal of the UAV group may become unjustifiably expensive. To transport objects to the target over relatively short distances, significantly cheaper UAVs can be used. For transportation, it is necessary to keep the UAV group within a specified flight corridor, prevent particle collisions with each other, and avoid areas dangerous for flight. If the number of aircraft is large, involving a group of qualified operators to control each aircraft is inefficient. This work proposes a new method of transporting a group by creating a virtual twin. A numerical analog of the UAV group is generated in virtual space. Control of each aircraft is carried out by transmitting control signals from a single operator's computer to the aircraft controllers. This program is implemented by simulating the motion of aircraft in virtual space in a nonlocal potential. When developing the flight scenario of the group, the flight corridor, initial velocities and coordinates of the aircraft, target coordinates, and the characteristic drift velocity toward the target are numerically described. The idea of the method is shown in Fig. 1.

In works [1, 2], a model of group motion in a three-dimensional flight corridor is implemented based on the numerical solution of a system of stochastic ordinary differential equations. A modernized method of integrating stochastic equations based on the Runge–Kutta algorithm is used. This integration method imposes increased requirements on computer hardware parameters. Therefore, the search for methods of integrating sys-

tems of equations of UAV dynamics based on more economical algorithms continues. In the present work, the Verlet algorithm, modernized for integrating the system of equations of dynamics of a UAV group taking into account a nonlocal potential that accounts for the repulsion of UAVs from the domain boundary and from each other, is employed. The algorithm is tested by comparing the results of numerical integration using the Verlet algorithm with the exact solution of a linear equation and a nonlinear dynamic equation integrated using the library program Wolfram Mathematica. The results of deterministic trajectories of UAVs inside a circle with intensive collisions of aircraft with walls and with each other are presented.

1. Description of the Verlet algorithm and test examples

1.1. Integration of the system of equations of motion

The equation of dynamics of a particle in a viscous medium taking into account an external force has the form

$$\frac{dV(t)}{dt} + \frac{1}{\tau}V(t) = F(X(t), t), \quad \frac{dX(t)}{dt} = V(t).$$

Here τ is the relaxation time, $F(X(t), t)$ is the external force.

The velocity Verlet integration algorithm consists of several steps [3].

First step

$$\begin{aligned} \frac{V(t + \Delta t/2) - V(t)}{\Delta t/2} + \frac{1}{\tau}V(t + \Delta t/2) &= F(X(t), t), \\ V\left(t + \frac{\Delta t}{2}\right) \left(1 + \frac{\Delta t}{2\tau}\right) &= V(t) + F(X(t), t) \frac{\Delta t}{2}. \end{aligned} \tag{1}$$

Second step

$$\begin{aligned} \frac{X(t + \Delta t) - X(t)}{\Delta t} &= V\left(t + \frac{\Delta t}{2}\right), \\ X(t + \Delta t) &= X(t) + V\left(t + \frac{\Delta t}{2}\right) \Delta t. \end{aligned} \tag{2}$$

Third step

$$\begin{aligned} \frac{V(t + \Delta t) - V\left(t + \frac{\Delta t}{2}\right)}{\Delta t/2} &= F(X(t + \Delta t), t + \Delta t), \\ V(t + \Delta t) \left(1 + \frac{\Delta t}{2\tau}\right) &= V\left(t + \frac{\Delta t}{2}\right) + F(X(t + \Delta t), t + \Delta t) \frac{\Delta t}{2}. \end{aligned} \tag{3}$$

For one-dimensional motion, the Verlet scheme is obtained as

$$\frac{dV}{dt} + \frac{1}{\Omega_E} V = F(X, t), \quad \frac{dX}{dt} = V.$$

The integration algorithm is written as

$$V(t + \Delta t/2) \left(1 + \frac{\Delta t}{2\Omega_E} \right) = V(t) + F(X(t), t) \frac{\Delta t}{2},$$

$$X(t + \Delta t) = X(t) + V\left(t + \frac{\Delta t}{2}\right) \Delta t,$$

$$V(t + \Delta t) \left(1 + \frac{\Delta t}{2\Omega_E} \right) = V\left(t + \frac{\Delta t}{2}\right) + F(X(t + \Delta t), t + \Delta t) \frac{\Delta t}{2}.$$

1.2. Testing the Verlet algorithm

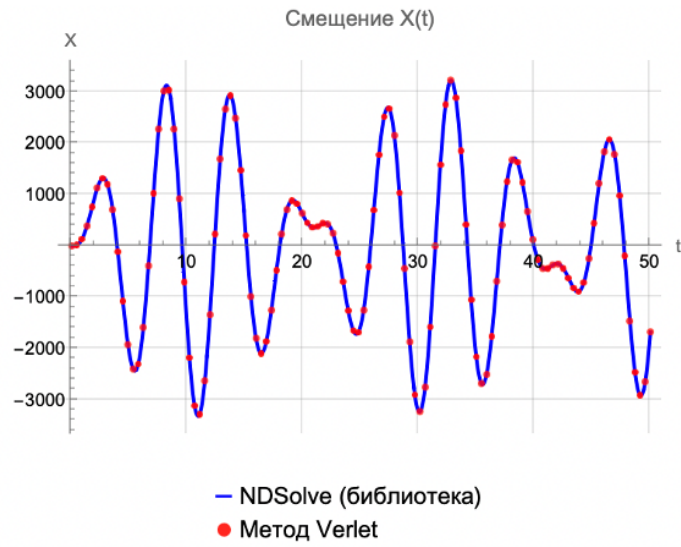
1.2.1. Oscillator with a periodic force

As the first test, an oscillator under the action of a periodic force is chosen. For this case, the numerical integration data are compared with the solution found using the library integration program.

The system of equations for the trajectory of an oscillator with a periodic force has the form

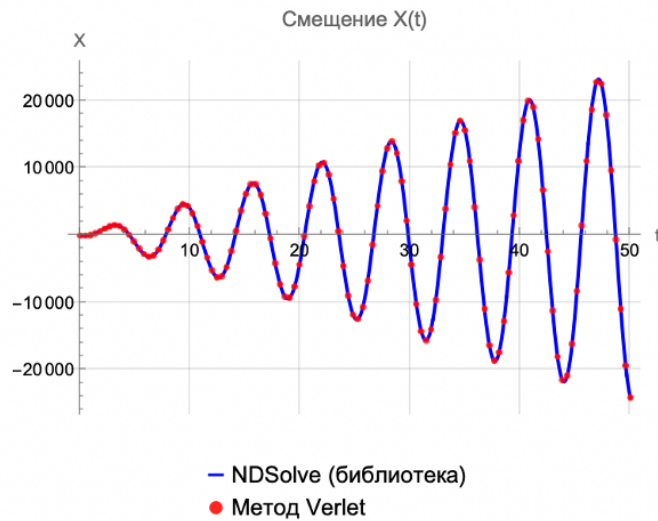
$$\frac{dV}{dt} + \frac{1}{\tau} V = -\omega_0^2 X + A_0 \sin(\nu t) = F(X, t), \quad \frac{dX}{dt} = V.$$

Figures 2 and 3 show a comparison of the results of calculating the oscillator motion using the Verlet method and the numerical solution using the NDSolve method from the library in the absence of resonance and in the case of frequencies close to resonance.



Параметры: $\omega_0 = 1$, $\gamma = 1.3$, $\tau = 1000$, $X_0 = 0$, $V_0 = 0$

Figure 1. Comparison of trajectory calculation results in the absence of resonance using the Verlet and NDSolve methods from the library



Параметры: $\omega_0 = 1$, $\gamma = 1.001$, $\tau = 1000$, $X_0 = 0$, $V_0 = 0$

Figure 2. Comparison of trajectory calculation results in the presence of resonance using the Verlet and NDSolve methods from the library

From Figures 2 and 3, it can be seen that for the linear equation the algorithm gives satisfactory results.

1.2.2. Mathieu equation

For the second test, the nonlinear oscillator equation with a frequency depending on the particle displacement is chosen

$$\frac{dV}{dt} + \frac{1}{\tau}V = \frac{1}{\Omega_E}A_0 \sin(vt) \sin\left(\frac{X}{2}\right) - \omega_0^2 X = F(X, t), \quad \frac{dX}{dt} = V.$$

In this case, a comparison is made between the numerical solution of the oscillator dynamics using the library integration program and the Verlet method.

Figures 4 and 5 show a comparison of the solution using the Verlet method and the numerical solution using the NDSolve method from the library.

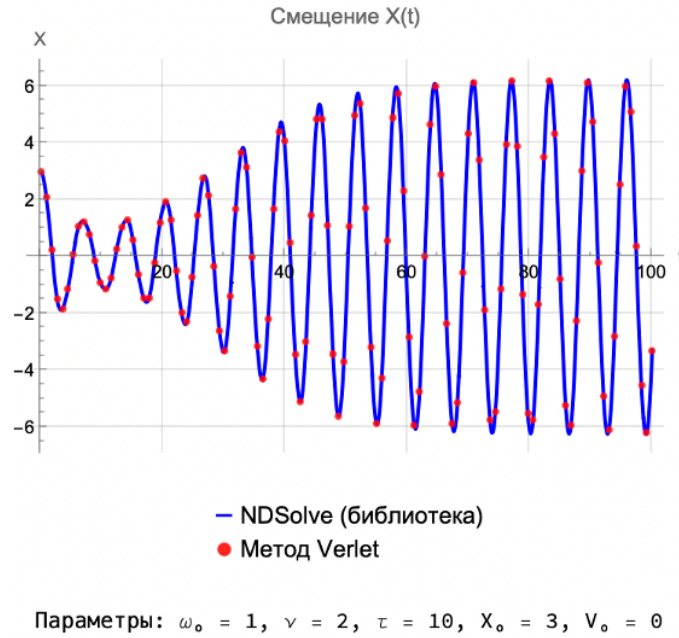


Figure 3. Comparison of trajectory calculation results in the absence of resonance using the Verlet and NDSolve methods from the library

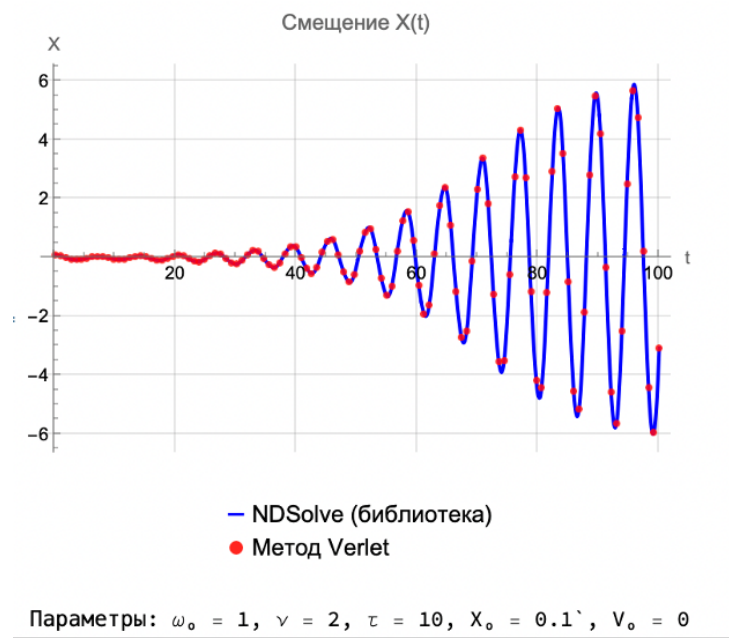


Figure 4. Comparison of trajectory calculation results at resonance using the Verlet and NDSolve methods from the library

As can be seen from Figs. 4 and 5, the Verlet algorithm leads to satisfactory results.

2. Dynamics of a UAV group in a confined volume

2.1. Equations of motion

The equations of motion of a group of particles taking into account dynamic relaxation have the form

$$\frac{dV^{(\alpha)}(t)}{dt} = \frac{1}{\tau^{(\alpha)}} (G^{(\alpha)} - V^{(\alpha)}(t)) + \frac{1}{m^{(\alpha)}} F^{(\alpha)}(X^{(\alpha)}(t), t), \quad (4)$$

$$\frac{dX^{(\alpha)}(t)}{dt} = V^{(\alpha)}(t). \quad (5)$$

Here $\tau^{(\alpha)}$ is the dynamic relaxation time of the UAV; $G^{(\alpha)}$ is the drift velocity toward the target, $m^{(\alpha)}$ is the mass of the UAV; $F^{(\alpha)}$ is the nonlocal force caused by all aircraft in the group and the repulsion of the aircraft from the wall.

2.2. Equations of motion in a nonlocal potential

The nonlocal potential leads to the motion of the group as a single system, in which the dynamics of each aircraft depends on the dynamics of all UAVs in the group taking into

account the impenetrability of the flight corridor. The equation of motion of an aircraft in a nonlocal potential has the form

$$\frac{dV^{(\alpha)}}{dt} = \frac{1}{\tau^{(\alpha)}} \left(\underbrace{G^{(\alpha)}}_{\text{drift velocity}} - V^{(\alpha)} \right) - \frac{1}{m^{(\alpha)}} \underbrace{\frac{\partial U_{\text{eff}}^{(\alpha)}}{\partial \Delta r} \frac{\partial \Delta r}{\partial r}}_{-F(X^{(\alpha)}, t)} \bigg|_{r=\Delta R^{(\alpha)}}, \quad \frac{dX^{(\alpha)}}{dt} = V^{(\alpha)}. \quad (6)$$

For this work, the drift velocity $G^{(\alpha)} = 0$; $\Delta R^{(\alpha)}$ is a vector directed from the surface of the aircraft along the shortest normal to the surface of the flight corridor or to the surface of another aircraft.

The effective nonlocal interaction potential consists of repulsive forces between aircraft and repulsion from the domain boundary

$$U_{\text{eff}}^{(\alpha)}(\Delta R) = \sum_{\beta \neq \alpha}^{N_0} U_{\text{rep}}(\Delta R^{(\alpha\beta)}) + \sum_{\text{wall}} U_{\text{wall}}(\Delta R_{\text{wall}}^{(\alpha)}). \quad (7)$$

The first term describes the interaction between aircraft, the second describes the confinement of the group within the flight corridor.

2.3. Potential of repulsion from the domain walls

UAVs are modeled as impenetrable spheres of radius $\delta^{(\alpha)}$. Figure 6 shows a diagram for calculating the repulsion force from the walls of the circle.

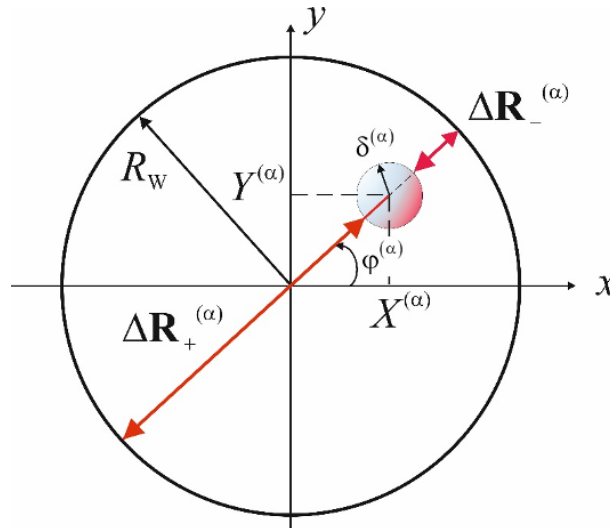


Figure 5. Diagram for calculating the repulsion force of an aircraft from the domain walls, R_W is the radius of the circle

The repulsion force is directed along the normal from the boundary surface to the nearest surface of the aircraft. In this case, these are $\Delta R_-^{(\alpha)}$ and $\Delta R_+^{(\alpha)}$. The radius vector of the aircraft is $R^{(\alpha)} = \{X^{(\alpha)}, Y^{(\alpha)}\}$.

The potential of repulsion of the aircraft from the domain walls is written as

$$U_W(\Delta R^{(\alpha)}, \delta^{(\alpha)}) = A_W \exp \left\{ \left(\frac{\delta^{(\alpha)}}{|\Delta R^{(\alpha)}|} \right)^{\sigma_W} - \left(\frac{|\Delta R^{(\alpha)}|}{\delta^{(\alpha)}} \right)^{\sigma_W} \right\}. \quad (8)$$

Here $\sigma_W > 0$ is the degree of localization of the potential near the boundary, $A_W > 0$ is the repulsion energy, J.

From (8) it can be seen that when approaching the surface $|\Delta R^{(\alpha)}| \rightarrow 0$ the potential exhibits super-exponential growth. As the distance from the boundary increases $\Delta R^{(\alpha)} = |\Delta R^{(\alpha)}| \gg \delta^{(\alpha)}$, the repulsion potential (8) tends to zero.

$$\begin{aligned} \frac{\partial U_W(\Delta R^{(\alpha)})}{\partial \Delta R^{(\alpha)}} = \\ - A_{\text{rep}}^{(W)} \frac{\sigma_W}{\Delta R^{(\alpha)}} \left\{ \left(\frac{\delta^{(\alpha)}}{\Delta R^{(\alpha)}} \right)^{\sigma_W} + \left(\frac{\Delta R^{(\alpha)}}{\delta^{(\alpha)}} \right)^{\sigma_W} \right\} \exp \left\{ \left(\frac{\delta^{(\alpha)}}{\Delta R^{(\alpha)}} \right)^{\sigma_W} - \left(\frac{\Delta R^{(\alpha)}}{\delta^{(\alpha)}} \right)^{\sigma_W} \right\}. \end{aligned} \quad (9)$$

Figure 8 shows a comparison of the result of differentiating the repulsion potential (8) with respect to relative distance. It can be seen that as the surface is approached, the repulsion force tends to infinity.

2.4. Potential of repulsion between two spheres

The scheme of repulsion of two UAVs from each other is shown in Fig. 8. The repulsion force is determined by the shortest distance between two spheres $\Delta R^{(\beta\alpha)}$, modeling two aircraft with radii $\delta^{(\alpha)}, \delta^{(\beta)}$.

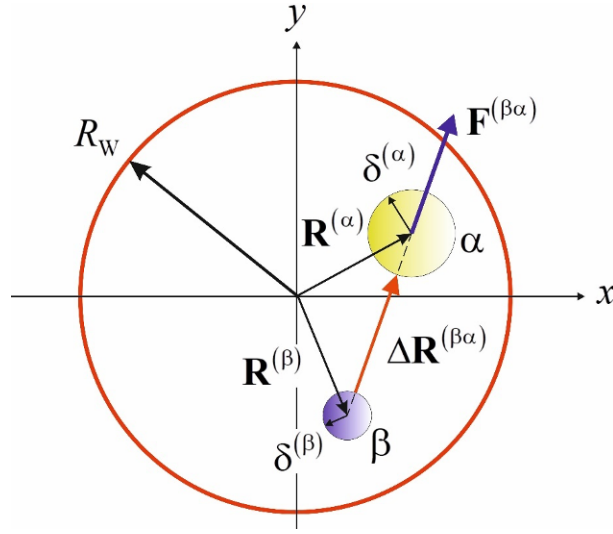


Figure 6. Scheme for calculating repulsion between two UAVs. $R^{(\alpha)}$, $R^{(\beta)}$ are the radius vectors of the aircraft

The repulsion potential has a form similar to the repulsion potential between an aircraft and the boundary (8)

$$U_{\alpha\beta}(\Delta R^{(\alpha\beta)}, \delta^{(\alpha\beta)}, \sigma_{\alpha\beta}) = A_{\alpha\beta} \exp \left\{ \left(\frac{\delta^{(\alpha\beta)}}{|\Delta R^{(\alpha\beta)}|} \right)^{\sigma_{\alpha\beta}} - \left(\frac{|\Delta R^{(\alpha\beta)}|}{\delta^{(\alpha\beta)}} \right)^{\sigma_{\alpha\beta}} \right\}. \quad (10)$$

Here $\sigma_{\alpha\beta} > 0$ is the degree of localization of the potential near the surface of the aircraft, $\delta^{(\alpha\beta)} = \delta^{(\alpha)} + \delta^{(\beta)}$.

From (10) it can be seen that when UAVs approach $|\Delta R^{(\alpha\beta)}| \rightarrow 0$ the repulsion potential increases super-exponentially, as the aircraft move away from each other $|\Delta R^{(\alpha\beta)}| \gg \delta^{(\alpha\beta)}$ the potential decays exponentially.

3. Algorithm for integrating the equations of motion of a UAV group

The algorithm for integrating the dynamics of a UAV group (4) – (10) is as follows. The transition from time layer t to layer $t + \Delta t$ is implemented according to the scheme described earlier.

First step

$$V^{(\alpha)} \left(t + \frac{\Delta t}{2} \right) = \left(1 + \frac{\Delta t}{2\tau^{(\alpha)}} \right)^{-1} \left\{ V^{(\alpha)}(t) + \left[\frac{1}{\tau^{(\alpha)}} G^{(\alpha)} + \frac{1}{m^{(\alpha)}} F(X^{(\alpha)}(t), t) \right] \frac{\Delta t}{2} \right\} \quad (11)$$

Second step

$$X^{(\alpha)}(t + \Delta t) = X^{(\alpha)}(t) + V^{(\alpha)}\left(t + \frac{\Delta t}{2}\right) \Delta t \quad (12)$$

Third step

$$V^{(\alpha)}(t + \Delta t) = \left(1 + \frac{\Delta t}{2\tau^{(\alpha)}}\right)^{-1} \times \left\{ V^{(\alpha)}\left(t + \frac{\Delta t}{2}\right) + \left[\frac{1}{\tau^{(\alpha)}} G^{(\alpha)} + \frac{1}{m^{(\alpha)}} F^{(\alpha)}(X^{(\alpha)}(t + \Delta t), t + \Delta t) \right] \frac{\Delta t}{2} \right\} \quad (13)$$

The integration step Δt is chosen empirically based on the stability criterion of the numerical scheme.

4. Calculation results

4.1. Motion of a single UAV in a circle

First, we present the results of modeling a single aircraft in a circle. The calculations were carried out in dimensional form. At the initial moment of time, the coordinates of the aircraft and the velocity components are chosen randomly. Figure 10 shows the trajectory of the UAV in a circle. It can be seen that the aircraft does not leave the domain boundary.

When the aircraft approaches the boundary, a repulsive force appears, changing the direction of motion (Fig. 10). We will call the sharp change in trajectory a "collision" with the wall. Note that the surfaces of the aircraft and the boundary of the flight domain do not come into contact.

The system of equations (6) describes motion taking into account velocity relaxation. The drift velocity is absent in this work, so over time the amplitude of the aircraft velocity tends to zero.

4.2. Motion of a UAV group

Next, the results of numerical integration of the dynamics of a group of 13 aircraft in a circle with impenetrable boundaries are presented. Figure 12 shows the initial coordinates of the aircraft, which were chosen randomly. A uniform random distribution was specified over the radial and angular variables. Figure 13 illustrates the initial distributions of the velocity components of the aircraft. The velocity components were chosen from a given range according to a uniform distribution. Figure 14 shows the trajectories of three arbitrarily chosen aircraft from the group. It can be seen that the UAVs do not leave the specified flight corridor. The sharp change in trajectories is associated with "collisions" of

aircraft with the boundary and with each other. There is no contact between the surfaces of the aircraft. Figure 15 shows the dynamics of the change in the axial velocity component of three selected aircraft. The sharp change in the direction of motion of the aircraft over time is associated with collisions of UAVs with each other and with the domain boundary.

Conclusion

The paper presents an original method for controlling a group of UAVs from a single operator's computer. A digital twin of the UAV group is created in virtual space. The operator sets the initial distribution of the aircraft in space and their initial velocities. The flight corridor is described. Subsequently, the motion of the group toward the mission target occurs autonomously. At the current moment in time, information about the position and velocity of each aircraft in the group is received by the operator's computer. At the next moment in time, the control signals to the aircraft controllers are generated as a result of solving a system of ordinary differential equations (ODEs) describing the dynamics of the aircraft in a virtual nonlocal potential, ensuring motion in a given flight corridor and the absence of collisions between aircraft.

The paper illustrates the efficiency of solving the system of ODEs using the Verlet method [3]. The Verlet method is tested on characteristic examples. The system of ODEs describing the dynamics of the group's motion in a virtual nonlocal potential is written down. A software implementation in Wolfram Mathematica is developed, the calculation results and their analysis are presented. The mathematical package CorelDraw was used in the design of the work.